	Independent field-scale measurement cadence	Observation opportunity (979 events)	Detection at 10 % FPR (held-out stations)	Field scale (≤ 200 m)	Cost / access	Main open risk
Coordinate today's fleet (S1C/D + NISAR)	median gap 2.0 d, longest 20 d OBSERVED	51.6 % OBSERVED	S1 34.3 %; NISAR HH 50.0 % OBSERVED	yes OBSERVED	free OBSERVED	looks miss short events OBSERVED
+ ROSE-L, 6 d (2 sats)	6 d, from ~2030 or later PLANNED	63.7 % SIMULATED	assumed as NISAR UNVALIDATED	yes PLANNED	free PLANNED	launch date uncertain PLANNED
Buy: Spire D-HSSM (100 m daily estimate)	daily estimate; S1 looks ~6 d beneath VENDOR CLAIM	UNKNOWN (product is daily) UNKNOWN	UNKNOWN — Stage 1 benchmark UNKNOWN	100 m, downscaled VENDOR CLAIM	price on request UNKNOWN	event timing never scored UNKNOWN
Fuse: SMAP (+ NISAR + S1)	SMAP 2–3 d at 9 km OBSERVED	76.6 % any quality; 26.7 % recommended OBSERVED	SMAP rec. 67.2 % (9 km) OBSERVED	no (9 km) OBSERVED	free + research ESTIMATE	fusion method untested on events UNVALIDATED
Build: ARMO, 1 satellite (C)	mean gap 1.8 d SIMULATED	77.2 % SIMULATED	target 70 % REQUIREMENT	100–200 m target REQUIREMENT	≈ US$ 111 M ± 40 % ESTIMATE	misses 80 % even if fully imaged SIMULATED
Build: ARMO, 2 satellites (D)	mean gap 1.4 d (longest 5.5) SIMULATED	81.5 % (362 km corridor imaged); 72.7 % (200 km strip) SIMULATED	target 70 % REQUIREMENT	100–200 m target REQUIREMENT	≈ US$ 150 M ± 40 % ESTIMATE	needs ~full-corridor imaging + 70 % retrieval UNVALIDATED

Recommendation (conditional): fund Stage 1 now. Proceed to ARMO Stage 2 only if no cheaper option meets ARMO's requirements on the same events (Gate 1, interim 2027; Gate 2, 2028), L-band retrieval reaches ≥ 70 % detection at ≤ 10 % false alarms (Gate 2), and Phase A shows the imaging swath, reflector and cost close (Gate 3). Otherwise buy, fuse, stop or redesign.

Evidence classes as labelled; dashed red = UNKNOWN (not zero). Not an economic Pareto frontier: costs are ± 40 % estimates and no option's benefit is monetised. Market facts dated 3 Nov 2026.